

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1449

South Colton 138 kV
May 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1449. The Interconnection Customer proposes a 297.1 MW solar photovoltaic facility interconnecting at the South Colton 138 kV substation in the Virginia area. The request was received on April 08, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 522 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,718,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Yaphank - Brush Tavern 138.0 kV	Clover - Bear 345.0 kV	117.0	311	17.6
Pinas 9403-9405 - S Colton Substation 138.0 kV	Brush Tavern - Rustburg 138.0 kV	102.3	236	14.7
Mitchell Dam - Skaggs Island Jct 2 345.0 kV line	Leavitt - Chancellor Row 345.0 kV	101.6	1,500	7.3
Pinas 9403-9405 - Evington Substation 138.0 kV line	Brush Tavern - Rustburg 138.0 kV L	100.7	231	14.7
Mitchell Dam - Haggin 345.0 kV	Leavitt - Chancellor Row 345.0 kV	100.5	1,228	6.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yaphank - Brush Tavern 138.0 kV	Rebuild line with higher-capacity conductor	\$4,687,000	31% (\$1,441,000)
Pinas 9403-9405 - S Colton Substation 138.0 kV	Reconductor line	\$2,271,000	31% (\$696,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mitchell Dam - Skaggs Island Junction 2 345.0 kV	Reconductor line	\$11,331,000	2% (\$205,000)
Pinas 9403-9405 - Evington Substation 138.0 kV line	Reconductor line	\$765,000	31% (\$234,000)
Mitchell Dam - Haggin 345.0 kV	Reconductor line	\$7,636,000	2% (\$142,000)
	Total allocated to project		\$2,718,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.